

Partisan Gerrymandering

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(talk at Brookline High School, 2/15/2011)

The Most Predictably Conflictual Issue

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- Suppose you work hard,

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- Suppose you work hard, graduate from BHS,

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- Suppose you work hard, graduate from BHS, go to a great college,

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- Suppose you work hard, graduate from BHS, go to a great college, get a graduate degree,

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~> Legislative Redistricting

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~> Legislative Redistricting

(Watch Carefully; We'll have a contest to redistrict Brookline shortly!)

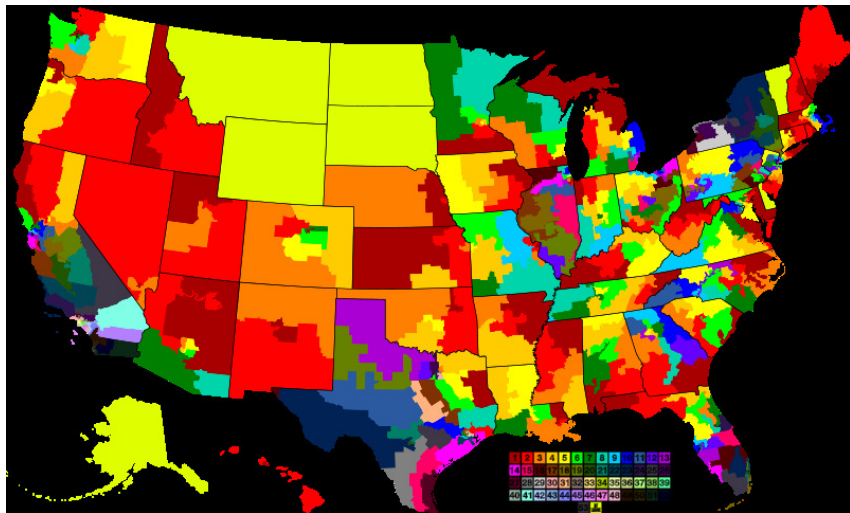
U.S. House of Representatives



Count Everyone $\rightsquigarrow$ Apportion Seats to States



Redistrict Within States



Fort Worth Star-Telegram

SEVENTEENTH TEXAS SENATE MEETING

Texas Senators Head For New Mexico To Block Redistricting

The Senators

Senators from all 31 Texas counties will head for New Mexico today to block redistricting.

By the way

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Redistricting Battle for Policy Analysis

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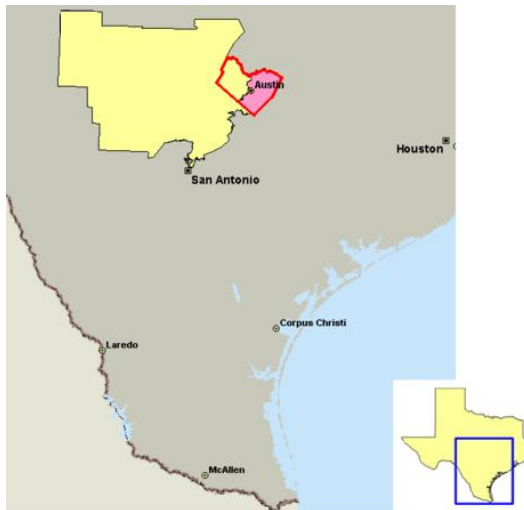
Redistricting

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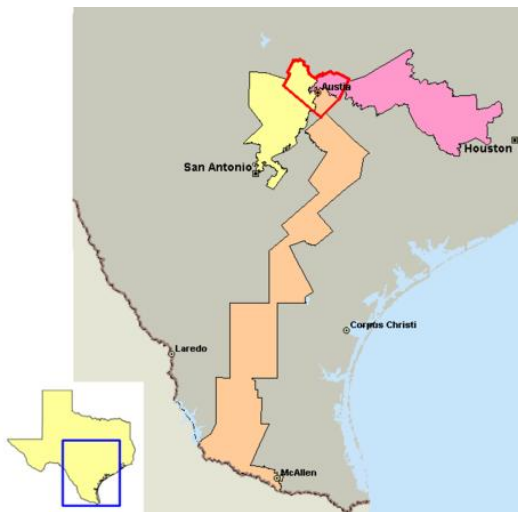
Is this fair?

Texas, *before* redistricting



Or is *this* fair?

Texas, *after* redistricting



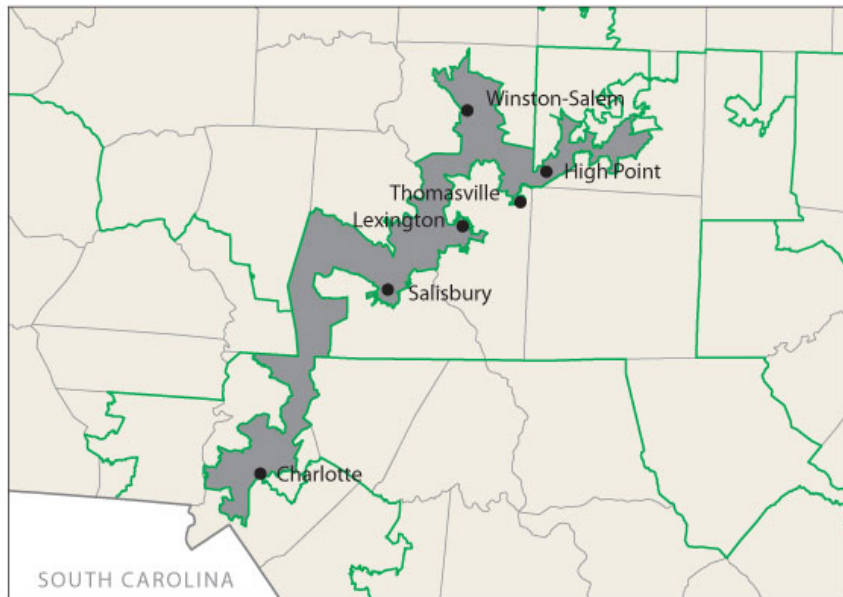
(Important but Inadequate!) Standards for Fairness

- 1 Equal population
- 2 Preserving “Communities of Interest”
- 3 Compactness (and contiguity)

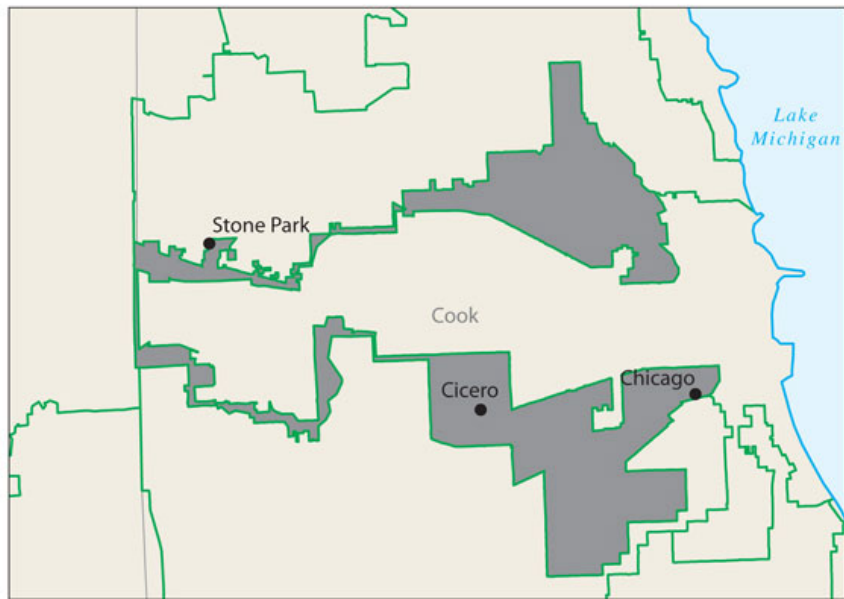
Arizona's 2nd Congressional District



North Carolina's 12th Congressional District



Illinois' 4th Congressional District



Governor Gerry's 1812 "Gerrymander" in Massachusetts



THE GERRY-MANDER.

How many (of 4) legislative seats should each party get?

Democratic vote % in each precinct

90	40	40	40
90	40	40	40
90	40	40	40
90	40	40	40

(Average = 52.5%)

A Republican Redistricting Plan

90	40	40	40
90	40	40	40
90	40	40	40
90	40	40	40

Average 90 40 40 40

⇒ Democrats win 1 of 4 seats

A Democratic Redistricting Plan

Average

90	40	40	40	52.5
90	40	40	40	52.5
90	40	40	40	52.5
90	40	40	40	52.5

⇒ Democrats win all 4 seats

New Standards for Fair Redistricting

- 1 Equal population
- 2 Preserving “Communities of Interest”
- 3 Compactness (and contiguity)
- 4 Partisan symmetry (no bias)
- 5 Electoral responsiveness

The Supreme Court on Partisan Symmetry

Cite as: 548 U. S. ____ (2006)

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Opinion of KENNEDY, J.

NOTICE: This opinion is subject to formal revision before publication in the preliminary print of the United States Reports. Readers are requested to notify the Reporter of Decisions, Supreme Court of the United States, Washington, D. C. 20543, of any typographical or other formal errors, in order that corrections may be made before the preliminary print goes to press.

SUPREME COURT OF THE UNITED STATES

Nos. 05–204, 05–254, 05–276 and 05–439

LEAGUE OF UNITED LATIN AMERICAN CITIZENS,
ET AL., APPELLANTS

05–204

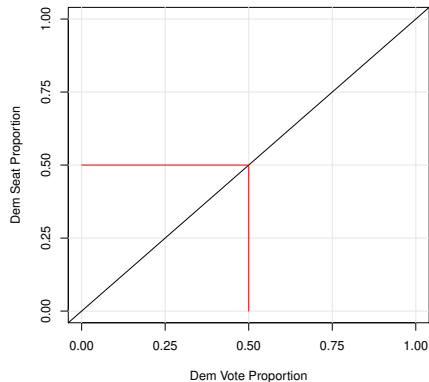
v.

RICK PERRY, GOVERNOR OF TEXAS, ET AL.

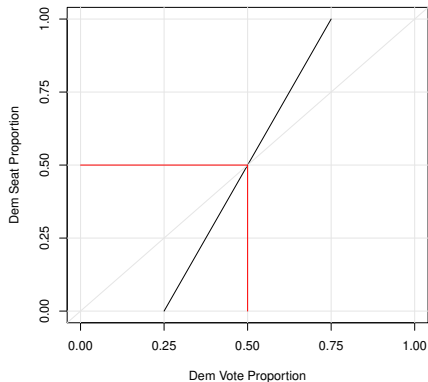
Fair Seats-Votes Curves

Partisan symmetry: When votes are split equally, seats are split equally

Fair, Proportional



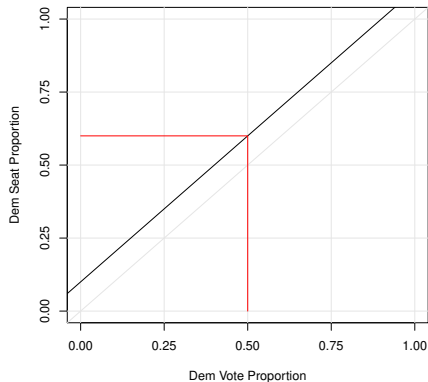
Fair, Responsive



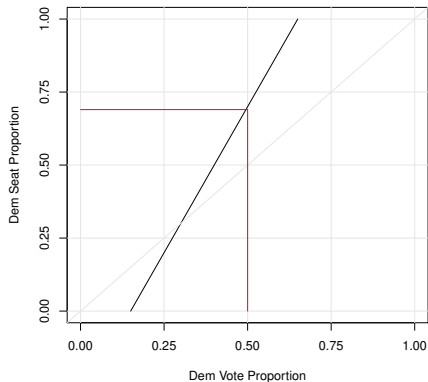
Biased Seats-Votes Curves

Partisan Bias: When votes are split equally, seats are split **unequally**

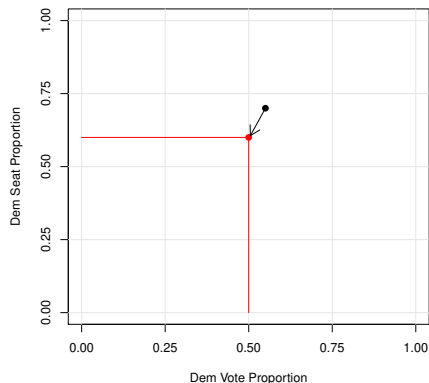
Biased, Proportional



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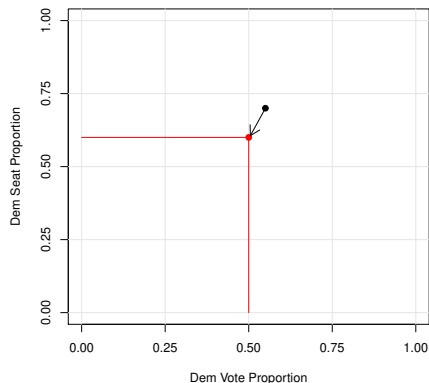


Calculating a Seats-Votes Curve

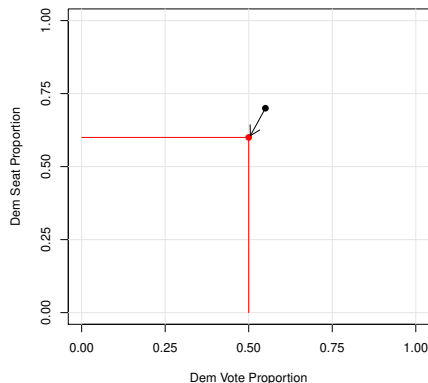


Calculating a Seats-Votes Curve

- Compute average vote and seat proportions; plot black dot

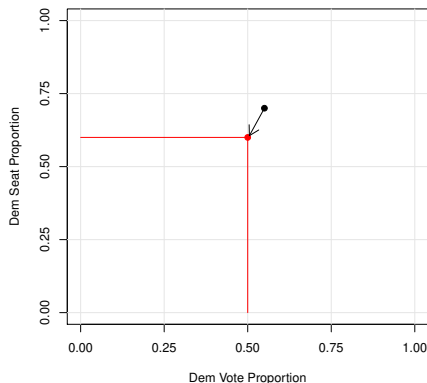


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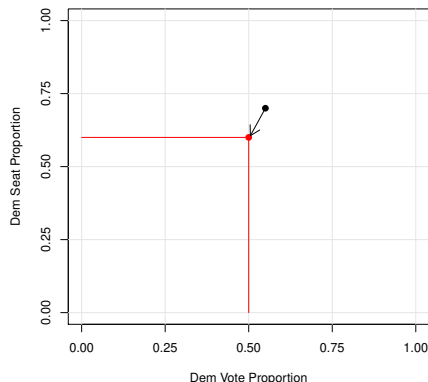
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- Find difference of vote average from 0.5

Calculating a Seats-Votes Curve



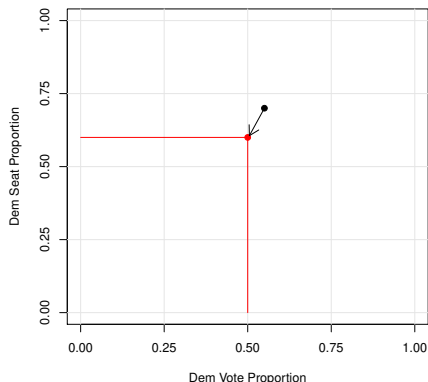
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- Calculate bias (difference in seat proportion from 0.5)

Brookline Precinct

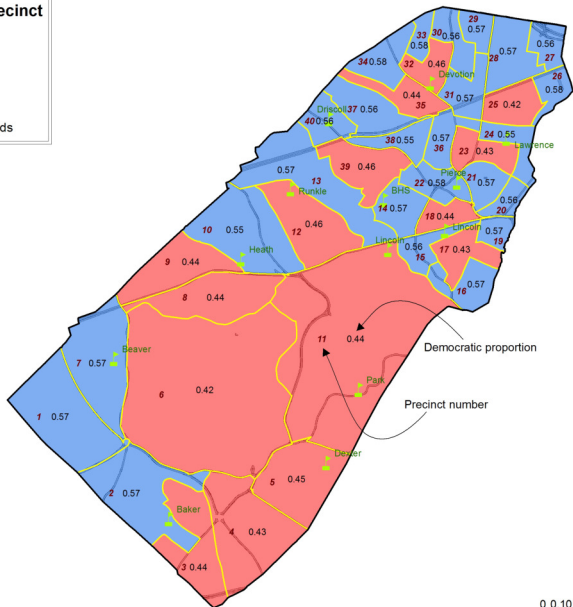
Winner

D

R

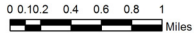
Schools

Major Roads



Democratic proportion

Precinct number



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- With partisan symmetry, math can make redistricting fair
- In practice, redistricting is a mess; law suits will be filed everywhere; rules will be broken; hundreds of millions of dollars will be spent
- To legislators, this is *the* most visible issue
- Redistricting is often invisible to the public. . . but not to you!